

Chapter 182

Audio Meters and Audio Monitoring

The Meters panel at the top of the Fairlight page provides a visual reference of the track levels in your mix, specialized meters showing the busses, meters for monitoring mix and loudness, and a video Viewer for seeing the current project as you work.

This chapter describes the use of these meters, and provides information about the different options that are available.

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The Monitoring Panel

Turning on the meters in the UI toolbar displays the Monitoring panel that runs along the top of the Fairlight page, which shows all of the audio meters that correspond to the tracks in the Timeline, as well as the Bus meters that correspond to the Mains, Subs, and Aux busses of your mix, the Control Room meters, and a video viewer.



The Monitoring panel

With the increased track counts necessary for immersive audio, DaVinci Resolve gives you options for extra metering so you can see more tracks in real time.

Right-clicking anywhere on the meters opens a menu that lets you choose between viewing them as a single-row or double-height Track display. This menu also allows you to choose between narrow or wide meters. When using the wide meter view, double-clicking anywhere on the Tracks meters panel switches between single-row and double-height view.

Hovering a mouse below the Tracks panel reveals a cursor that lets you change the height of the Monitoring panel.

The Compound meters and the Bus Output meters can also be resized by hovering on the right side of the Loudness meter and dragging to the desired dimensions.

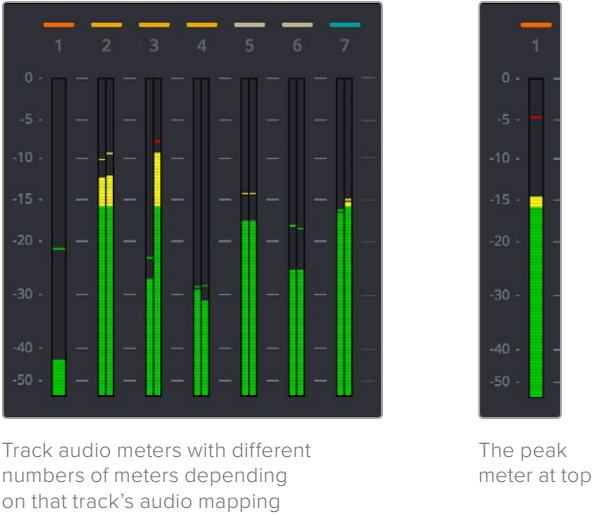


The double-height Monitoring panel

Track Meters and Monitoring Controls

At left, a row of audio meters corresponds to the channel strips of the Mixer, one meter for every audio track in the timeline. Each track displays the number of meters that corresponds to that track's audio mapping, with mono tracks having a single audio meter, stereo tracks having two, 5.1 tracks having six, and so on.

Each track and bus meter (with the exception of the Loudness meters) display RMS (root mean square) levels against a dB scale. A single line indicating the maximum value at any given moment in time is held briefly just above the current RMS levels, which appear as a solid bar extending from the bottom of the meter. RMS meters display a weighted “average” representation of the audio level that’s closer to the way audio is actually perceived, although not as accurate in measuring perceived loudness as the loudness meters discussed later in this section.



Each meter bar is color coded to indicate three different sound level “zones,” from low to medium (green) to moderately high (yellow) to very high (red).

Each meter is identified by the track number it represents (track names are not shown over track meters) as well as the color of that track.

Bus Meters

To the right of the track meters are the bus meters, in which all user-created mains and busses appear, separated by type, and each displaying the number of meters that corresponds to that track’s audio mapping. These meters allow you to monitor the sum of all tracks that have been routed to a particular bus, as you can on bus channel strips.

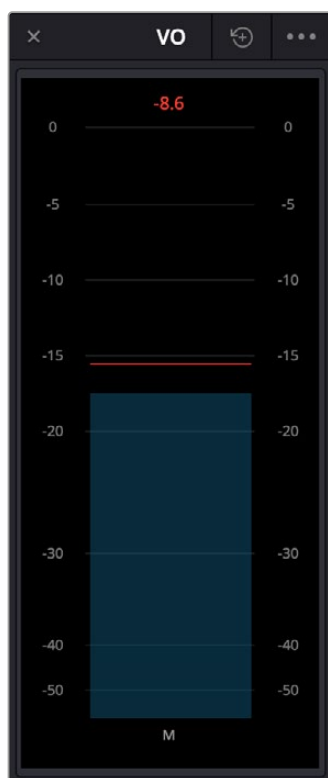


Meter Plugin

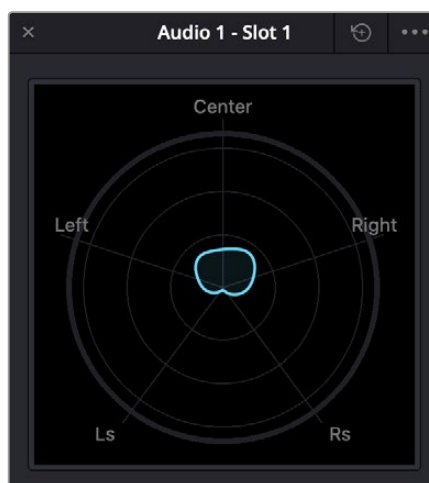
There is a Meter plugin available for temporarily adding a meter to a specific track, FlexBus, or when using fixed busing to Sub, Aux, or Main. These are sample peak processing meters that are useful for instances where you want a large meter that focuses on a specific bus while you're working.

These meters are presented very simply, with a gray bar indicating level and a red peak line that holds for two seconds, which indicates the highest peak. A numeric reading at the top of the meter gives the exact level, in dB. This number continues to hold, indicating the loudest level measured for any given stretch of playback. The option menu in this meter's floating window presents different settings you can choose.

For more information, see *Chapter 181, "Fairlight FX."*



The Meter Fairlight FX



The signal here is radiating more to the right, indicating the panning of the audio.

Surround Analyzer

The Surround Analyzer is a graphical meter that shows a spatial image of the audio being measured, rather than a typical bar graph meter. Due to its changing shape because of the signal being played, sometimes it is referred to as the “jellyfish meter.”

This type of metering is very useful; rather than relying on bars to indicate the directions in which audio is radiating, you can clearly see the relationship of all of the channels to one another.

Compound Meters and Output Bus Selection Menu

The compound meters, to the right of the bus meters, consist of Control Room meters and Loudness meters, shown side-by-side to provide a comprehensive analysis of your overall audio mix. Below, dropdown menus let you choose which bus you want to monitor, as well as which set of speakers you want to use to do the monitoring.

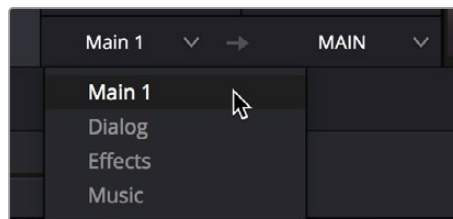


The Control Room meters (at left) and Loudness meters (at right)

Monitoring Menus

Bus Monitoring Menu

The bus monitoring dropdown menu determines which bus is monitored by your audio output (speakers or headphones), and thus which gets analyzed by the compound meters, letting you choose which bus you want to monitor as you work. You can choose one of your mains or any other bus (or track) you've set up that you want to focus on for more detailed work.



The monitoring menu lets you choose which bus you want to monitor while working.

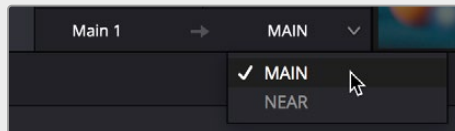
Speaker Set Monitoring Menu

To the right of the Monitoring menu, the output is set to "Auto" by default (if you have Automatic speaker configuration checked on in Preferences > Video and Audio I/O > Audio I/O). This means that your basic speaker set up is routed to the stereo or surround output your hardware supports automatically and can't be altered.

Monitoring with Multiple Speaker Sets

If you want to manually configure multiple speaker sets, unchecking the automatic speaker configuration preference will display a dropdown to the right and lets you choose one of the available sets of speakers once they have been configured in the Video and Audio I/O panel of the System Preferences. This gives you the flexibility to quickly compare your mix on a variety of speakers and configurations to see how it holds up in different situations.

For more information on configuring different speaker setups, see *Chapter 4, “System and User Preferences.”*



The monitoring menu lets you choose which bus you want to monitor while working.

Control Room Meters

The mustard-colored Control Room audio meters show the sum of all audio channels that are routed to the currently selected bus being monitored (as selected in the dropdown menu below). These are peak meters measured in dBFS.

A true peak audio measurement is displayed at the top of the Control Room meter.

Loudness Meters

The set of meters all the way to the right are the Loudness meters, which consist of a set of two graphical meters and a numerical readout. This lets you keep track of the “integrated loudness” of the overall mix, which is the standard that all contemporary mixing specifications refer to when specifying client deliverables. Unlike the RMS audio meters found in the Timeline or Mixer, which measure audio in dB, Loudness meters do a different kind of analysis, measured in LUFS (loudness units relative to full scale).

What Is LUFS?

LUFS measurements evaluate sound levels that are averaged over time referenced to full scale. Full scale is the loudest level a digital audio system can handle before overloading (“clipping”). LUFS allows totally different types of program material to be matched in perceived level by human ears and has become the standard way that mixes are measured for final delivery of your program material. LUFS are shown as negative numbers, such as -23, -14, etc., because the level is referenced to full scale (digital 0).

Loudness Meter Options

Two options in the General Options of the Project Settings let you customize the Loudness meters.

- **Target Loudness level:** Lets you set the LUFS value that's used as a reference level for loudness metering. Defaults to –23 LUFS, which conveniently makes the display of these meters scale similarly to traditional audio meters that you're already used to.
- **Loudness Scale:** Lets you choose which scale you want to use with which to measure the meters. Options currently include the default of EBU +9 Scale (–18 to +9), and EBU +18 Scale (–36 to +18).

Support for Multiple Loudness Standards

The Loudness Meter can be switched among a variety of international industry-standard loudness monitoring standards. The standard you choose uses the integrated loudness value (along with a specified tolerance defined by each selected standard) to indicate whether or not the current mix level is of acceptable loudness via color coding of the Integrated Loudness value, and in the Integrated Loudness graph described below. Blue values indicate loudness levels that are below tolerance, yellow indicates loudness values that are within tolerance, and red indicates loudness values that are above tolerance.

The built-in standards you can switch among include the following:

BS.1770-1: An older loudness standard used by DaVinci Resolve version 15 and before.

BS.1770-4: The most up-to-date loudness standard as of DaVinci Resolve 16; the algorithms specified by this standard govern the other standards that are listed below in this dropdown menu.

ATSC A/85: The American standard for acceptable loudness in broadcast.

EBU R128: The European standard for acceptable loudness in broadcast.

OP-59: The New Zealand and Australian standard for acceptable loudness in broadcast.

TR-B32: The Japanese standard for acceptable loudness in broadcast.

AGCOM 219: The Italian standard for acceptable loudness in broadcast.

NETFLIX: The Netflix standard for acceptable loudness in broadcast.

YouTube: The YouTube standard for acceptable loudness in broadcast.

Disney 2.0: The Disney 2.0 standard for acceptable loudness in broadcast.

Disney 5.1: The Disney 5.1 standard for acceptable loudness in broadcast.

All of these loudness standards are available for off-line readings as well, using the Loudness Analyzer described in the next section.

NOTE: The target peak meter now uses the BS.1770-4 standard for measuring maximum “true peak,” which means that this meter is capable of measuring “inter-sample peaks,” rather than only the peaks at each sample of a waveform.

Graphical Loudness Meters

Two separate meters give you a dynamic graphical measurement of the loudness of the selected bus being monitored according to the loudness standard you've selected, which determines how to analyze the subjective loudness of a given audio mix for purposes of compliance with required broadcast quality control (QC) standards.

- A steel-blue meter labeled M (for momentary) has as many channels as the selected bus you're monitoring, excluding the LFE channel(s) of surround formats, which aren't factored into loudness metering. This meter measures LEQ (equivalent sound level), within a 400ms window following the playhead as measured every 100ms. This lets you evaluate the LUFS (Loudness Units Full Scale) level of the mix at the current frame as you play. This discrete-channel analysis is used to calculate all other values of the loudness metering system.
- A second steel-blue mono meter to the right displays the sum of all channels in the M meter, displayed in LU (loudness units). The number value displayed at the top of this meter is the maximum LU value that's been analyzed during any stretch of timeline playback. This value is held until it's reset, either by stopping and initiating playback a second time while Link to Playhead is enabled, or when you click the Reset button at the bottom of the loudness meter area.

Numeric Loudness Meters

A set of values to the right of the meters give running reports on the audio level of your mix. While the graphical meters are useful for analyzing your mix as you work, these numeric readouts are particularly valuable for providing the strict information you need to adhere to written QC standards. Their meaning is as follows:

Short: Measures the average LU level over a 30-second window following the playhead.

Short Max: Shows the maximum level over the same 30-second window. This analysis is required by EBU R128.

Range: Measures the dynamic range of the Loudness of your mix (in LU, or loudness units), which is the difference between the average soft and average loud parts of your mix. Analyzes the overall loudness over a played range of the mix, discounts the lowest 10% and highest 5%, and then gives a standardized expression of the difference between the remaining soft and loud levels that were analyzed. The window of analysis is as long as you've been playing. This analysis is required by most QC specifications.

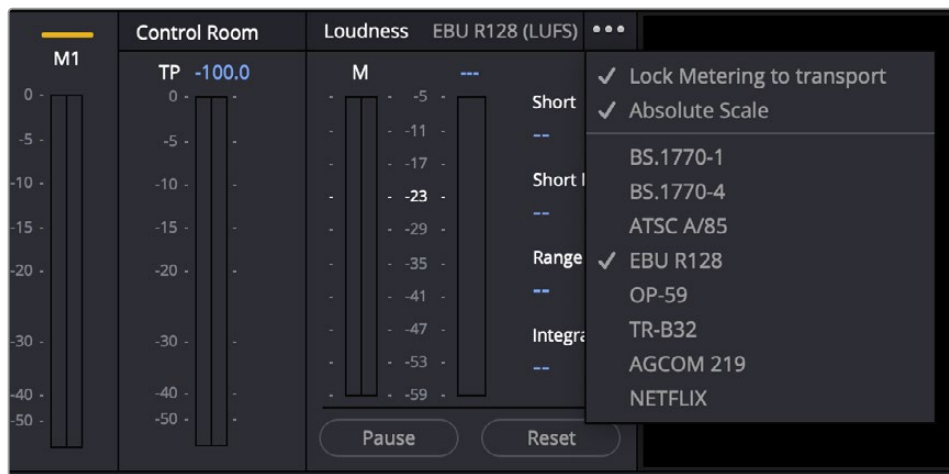
Integrated: Measures the LUFS value of the portion of the range of the mix you've played through. As you play, this integrated value accumulates. This analysis is required by most QC specifications.

Absolute Scales and Dialog

While some users prefer to measure their levels to correspond to a relative scale of "0," similar to a VU meter where the needle rides above the "0," others want to see the absolute measure of the amplitude in LUFS and true peak. By default, the Loudness meter is set to relative scale, but you now have the option to choose between relative scale and absolute scale in the Loudness meter.

Relative scale in the Loudness menu is relative to the selected scale, so a loudness unit of 0 corresponds to the target of the chosen measure type. For instance, if EBU R128 is selected, whose target measure is -23dB LUFS, the "0" LU (Loudness Unit) is -23dB. If ATSC A/85 is chosen, whose target is -24dB, then that becomes the corresponding equivalent of the relative LU of 0.

When using the absolute scale, the Loudness meter displays the increments to reflect the chosen measure type. In absolute scale the EBU R128 meter will display -23 instead of the relative scale's 0.



The option in the Loudness panel reveals the various measure types as well as the option for absolute scale.

Using the Loudness Meters

When using the Loudness meters to do a structured analysis of your mix to determine QC adherence, a group of controls let you determine when analysis begins and ends.

Lock Metering to Transport: This setting is found in the compound meter option menu. When enabled, all loudness metering analysis is automatically reset whenever you move the playhead to another location in the Timeline. This is useful when you're spot-checking different parts of your mix, or working on a particular scene. Uncheck this option if you want the measurement of playback to that point in the Timeline to remain.

Absolute Scale: An absolute measure of the selected scale type.

Pause and Reset buttons: When you're doing a formal analysis of your mix, the Reset button lets you reset all currently accumulated analyses, and the Start button initiates loudness value accumulation. If you need to stop playback briefly to do something else, you can click Pause, and then click Resume when you're ready to continue the analysis.

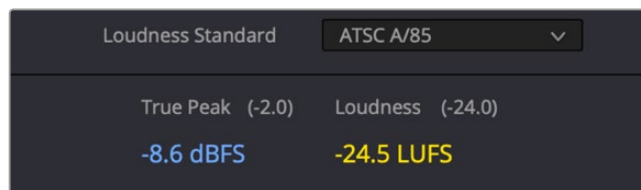
Offline Loudness Analyzer

Users also have the ability to analyze an audio file's loudness offline. This is a fast way to measure loudness in imported audio files or bounced mixes.

To initiate offline loudness analysis:

- Right-click the file in the Timeline, and choose Analyze Audio Level from the contextual menu. A dialog box displays the available options for measurement.

All of the measurements available in the real-time loudness meter are accessible in the dropdown menu inside the Analyze Audio Level panel. When the analyzer mode is chosen, the target measurements for that mode will display next to the reading in parenthesis. Once you have clicked the analyze button, the results will display in the panel next to the chosen mode's target measurement values.



Select the audio file, choose the desired analyzer mode, in this case EBU R128, and click analyze.

Visualizing Loudness Monitoring

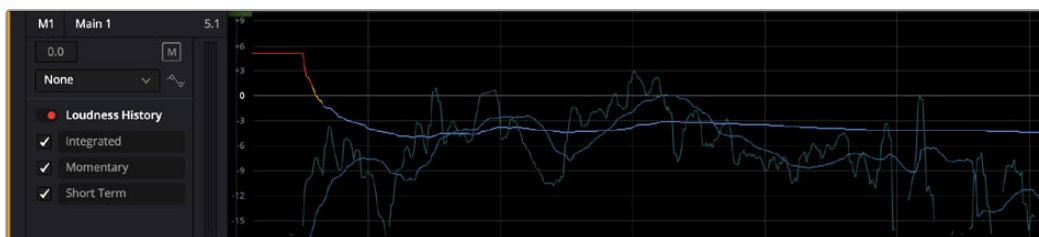
When you show the track of a Main bus in the Timeline, as long as the track is high enough (ahem, tall enough), you can show or hide a series of “Loudness History” curves to visualize the loudness analysis of the mix on that bus over the duration of the mix.

The available curves are:

- **Integrated:** A thick curve shows the averaged “integrated” loudness analysis of the current mix, which is a measurement that’s taken from the beginning to the end of playback. This graph is the primary gauge of whether or not measured loudness is acceptable. The color of each segment of this curve indicates whether that part of the mix is “to spec.” Blue indicate loudness levels that are below tolerance, yellow indicates loudness values that are within tolerance, and red indicates loudness values that are above tolerance. By evaluating the colors of the curve, you can easily spot which parts of your mix might need adjustment to meet the necessary specification.
- **Momentary:** A measurement of the loudness measured in the past 400ms, shown by a thin green-blue curve, which provides an analysis of transient level changes.
- **Short Term:** A measurement of the loudness of the past 3 seconds, shown by a thin blue line, which provides a more averaged analysis than the Momentary curve, yet still indicates the dynamics of the mix.

To show the loudness history for Main 1:

- 1 Open the Automation controls by clicking the automation button on the Fairlight toolbar.
- 2 Open the Index, and click the eye button for the Main you want to see in the Timeline. Loudness History appears as an option in the track header controls as long as the track is tall enough to show the controls. These controls will be hidden on short tracks.
- 3 Turn on the Loudness History toggle, and check the curves you want to see. The Integrated, Momentary, and Short Term loudness analyses can be individually displayed or hidden, to expose overlapping graphs in that track in which you can see your program’s loudness over time.



Viewing the loudness graph of the mix going out of a Main

NOTE: Currently, loudness history is only supported for Main 1.

Metering Options

DaVinci Resolve has a variety of metering options that allow you to tailor metering to your workflow. The settings appear in Project Preferences > Fairlight on the Audio Metering pane, but they affect mixer, Fairlight effects, or master metering on the Cut and Edit pages.

Level Metering Options

You can choose the response characteristic of the level meters for channel strips and Fairlight FX.

The Meter Type dropdown allows you to choose between IEC 60268-18, Digital VU, and Custom response characteristics. Both meter types have separate “hold and fall” metering, allowing you to see the highest peak that is reached.

IEC 60268-18: A digital PPM-type meter with reference standard of -18 dBFS, a fast response to peaks, and a slower release characteristic. This is the default in DaVinci Resolve and is used in all Blackmagic Design software and hardware products.

Digital VU: A dual-value meter, showing the peak level as a single segment with fast ballistics, and the RMS (volume unit) as a bar graph. It has a far faster quasi linear decay characteristic, making it easy to monitor average levels while allowing sounds with quick transients and decays to be more easily tracked. This option can be best for audio editing and mixing.

Custom: Allows each aspect of the meter response to be chosen, including level detector, scale, peak, decay and peak indication.

Pre-fader Metering

By default, “Pre fader metering on tracks” is unchecked (disabled), setting the metering on channel strips to be post (after) the fader level and mute button. For example, if the source audio file has peaks that are hitting -2 dBFS, and the channel fader is lowered from 0 to -10, then the peaks that show on the meter will now be at -12.

When pre-fader metering is on, the metering point is set to be before the fader, and is not affected by the fader’s position. However, the meter is affected by clip gain settings or key framing as these level changes occur before the clip’s signal enters the mixer’s signal path.

With pre-fader metering, you could lower the fader to be fully off (minus infinity) and the meter will still read the level of the source signal. This allows you to always be aware of the source signal and what it is doing before being altered by mixer’s controls or processing.

Many video editors prefer to work with post-fader metering, but most audio mixers prefer to work with pre fader in order to be aware of what the actual source level is at any time.

Solo and Pre-fader Metering

When pre-fader metering is enabled, soloing one or more tracks will show a lighter “ghost” version of meter activity on non-soloed tracks, allowing you to always see audio levels on tracks that are muted by the soloing process. This makes it easy to focus on tracks that are actually audible while still seeing activity on those that are not.

If pre-fader metering is off (the default), when one or more tracks are soloed, no metering occurs on tracks unsoloed tracks.



Pre-fader metering with light shaded meters on non-soloed tracks

Target Loudness Level

Allows the target loudness level in LUFS for the master Loudness meters to be set to your desired output target level. For example, DaVinci Resolve’s default loudness standard target is -23 LUFS, but the YouTube target LUFS specification is -14 LUFS. If -14 LUFS is set, the “0” mark on the loudness meter scale moves to this level, allowing you to focus your master mix levels towards that loudness standard.

Viewer

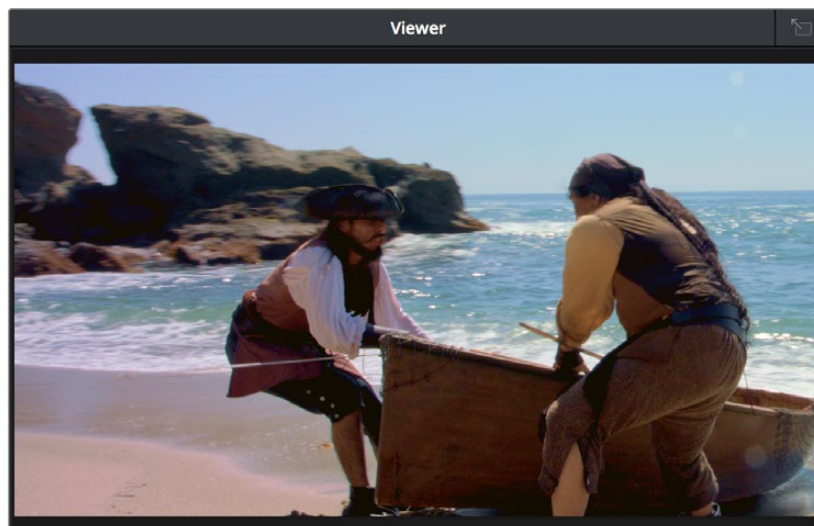
A small viewer to the right of the Monitoring panel shows the frame of video at the position of the playhead. This is the same image that's output to the external broadcast display of your workstation if you have one connected.



The Viewer lets you see the picture you're mixing to.

Clicking the Expand Viewer button at the bottom right-hand corner lets you open the Viewer into a floating window, which you can then position anywhere you want.

To close the floating Viewer, click the Dock Viewer button at the upper right-hand corner of the floating viewer window.



Click the button at the upper right-hand corner to dock the Viewer again.

Cinema Viewer on Fairlight Page

Sometimes when you need to verify lip sync, sound effects sync, or simply review a new section of a mix you've been working on, it helps to watch the visuals of the program full screen as you listen to your mix. You can now set the Fairlight page Viewer to Cinema Mode by choosing Workspace > Viewer Mode > Cinema Mode (Command-F).